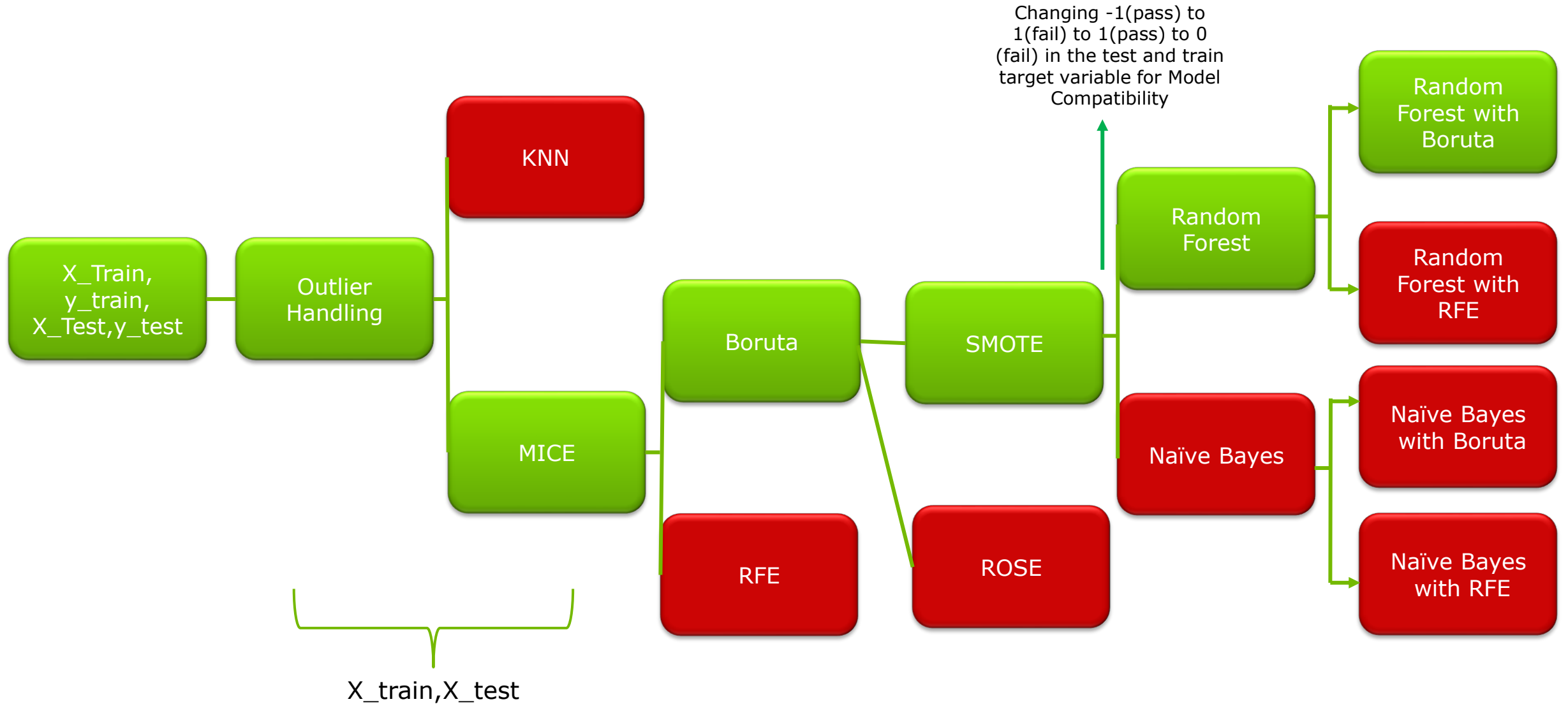


Agenda

1. RECAP
2. Scaling
3. Modelling Comparison
 - Random Forest
 - Decision Trees
 - K-Nearest Neighbors (KNN)
 - SVM
 - Gradient Boosting Classifier
5. Grid Search
6. Model Comparison
7. Summary & Best Practices

Flow Chart

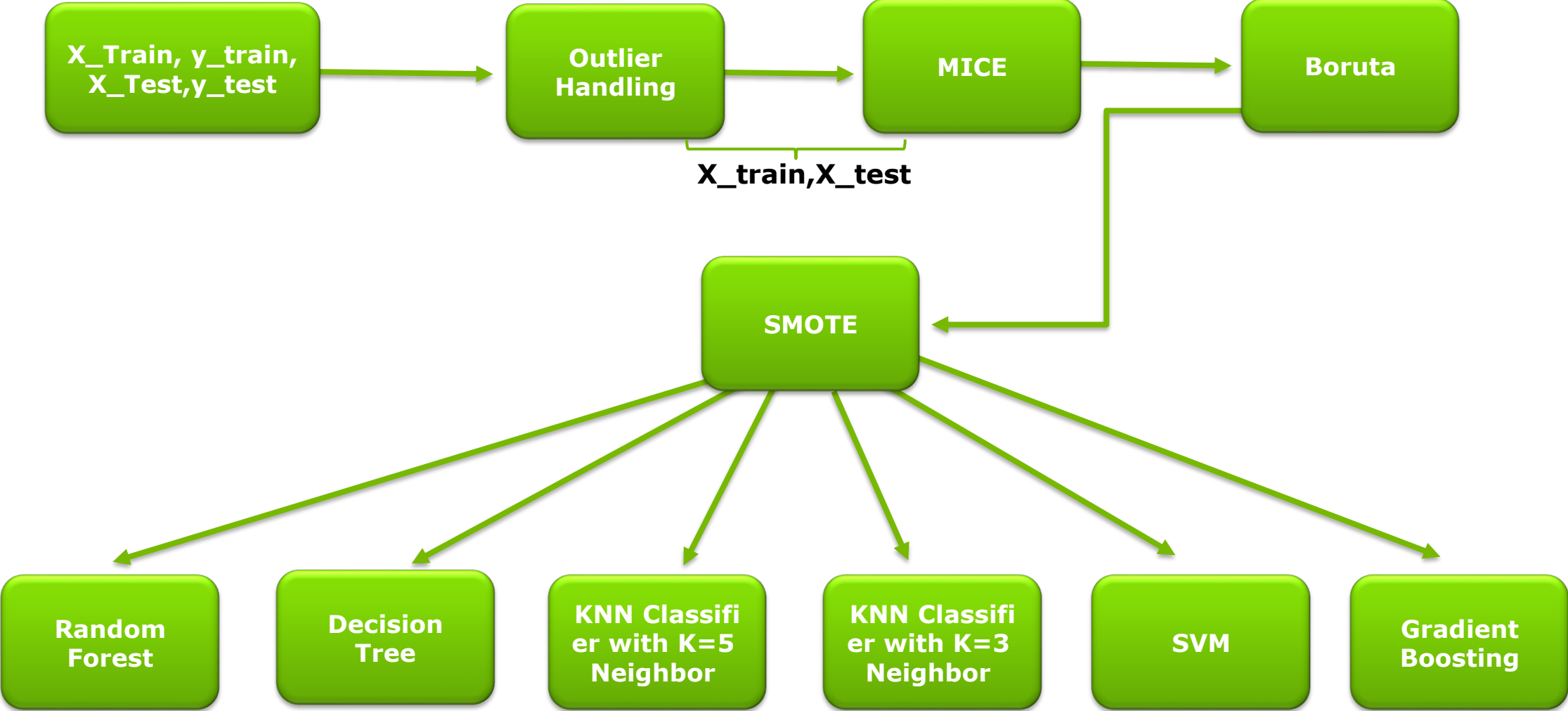


Scaling : Min & Max Scaling

K Nearest Neighbors	Support Vector Machines	Gradient Boosting Classifier
Distance Sensitivity	Optimal Hyperplane Calculation	Gradient Descent Optimization
1. Equal Contribution 2. Faster Convergence 3. Improved Performance		

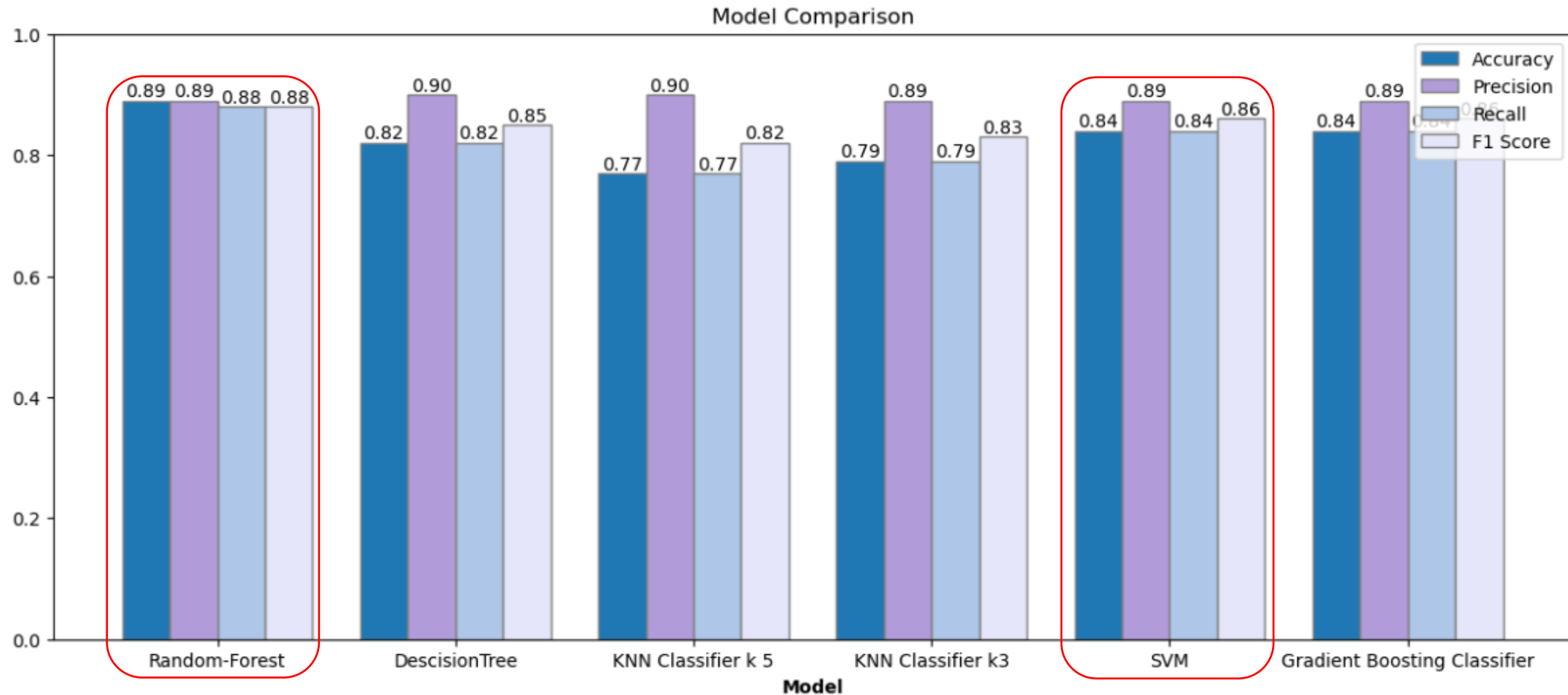
Flow Chart Attempt 1

Changing -1(pass) to
1(fail) to 1(pass) to 0
(fail) in the test and
train target variable for
Model Compatibility

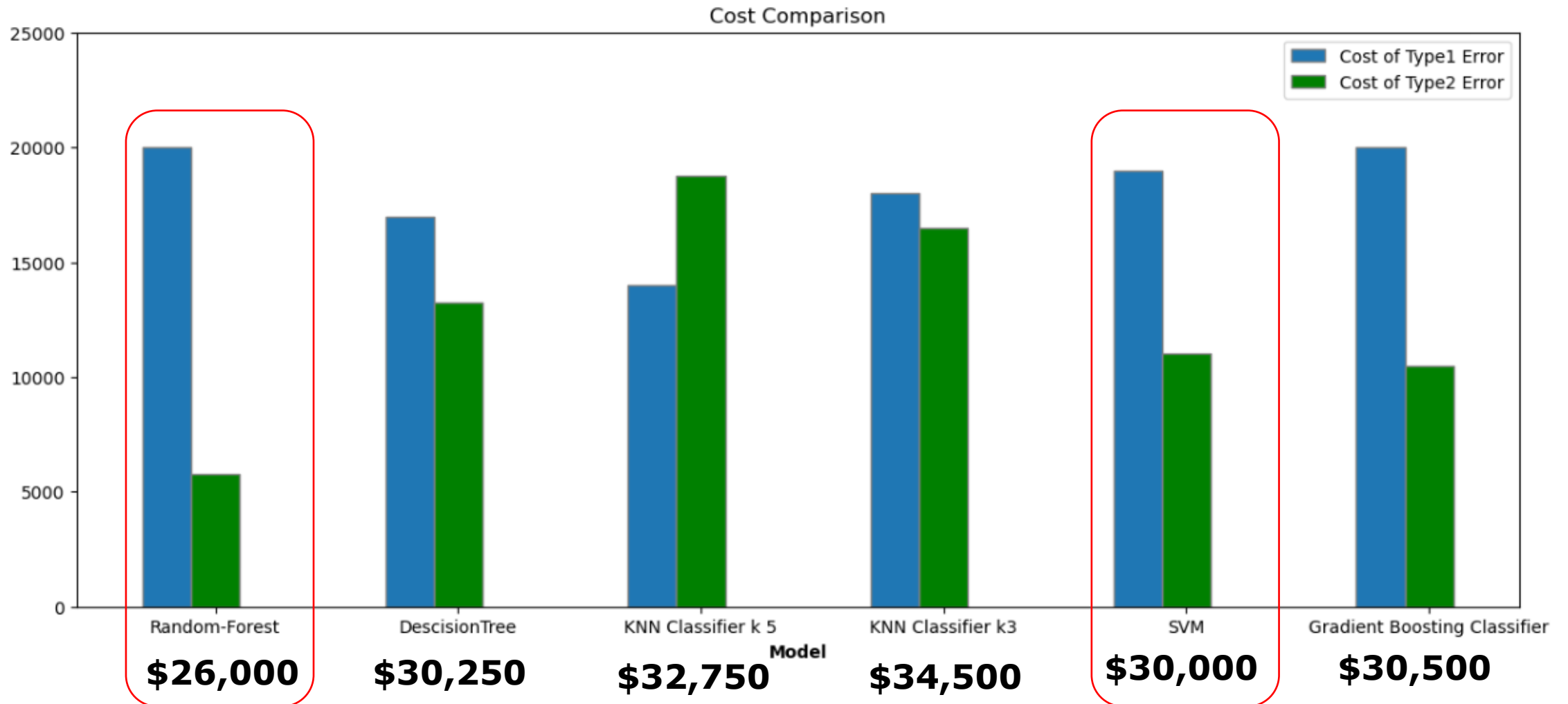


Modelling Comparison

Measures of goodness of fit



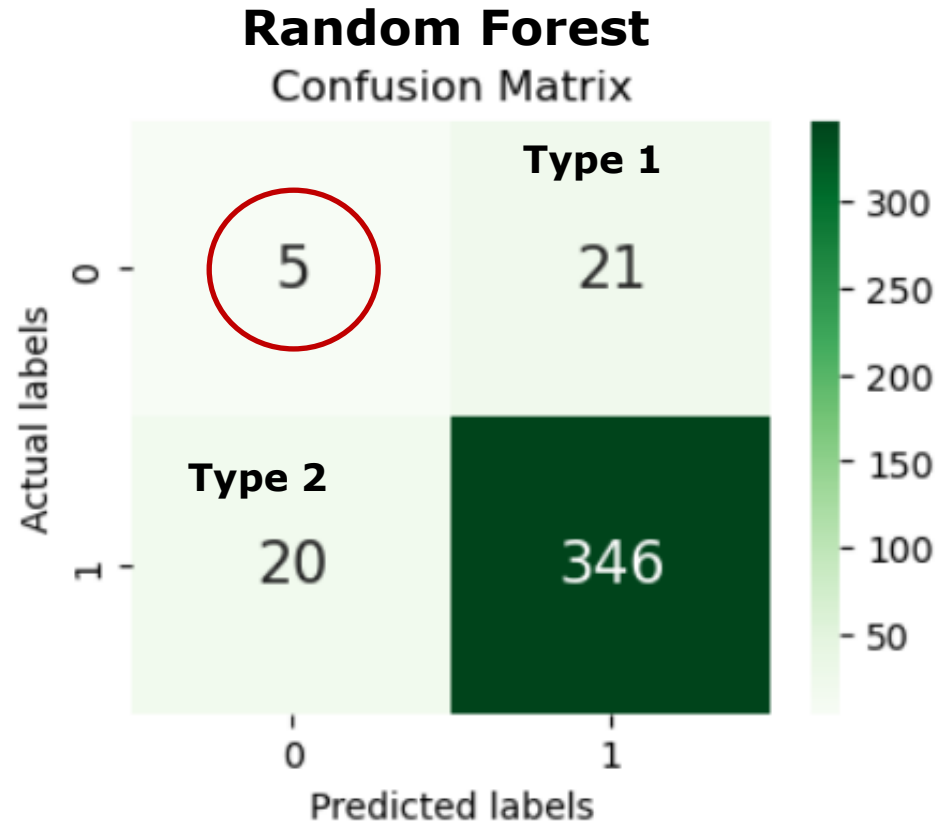
Costs Comparison



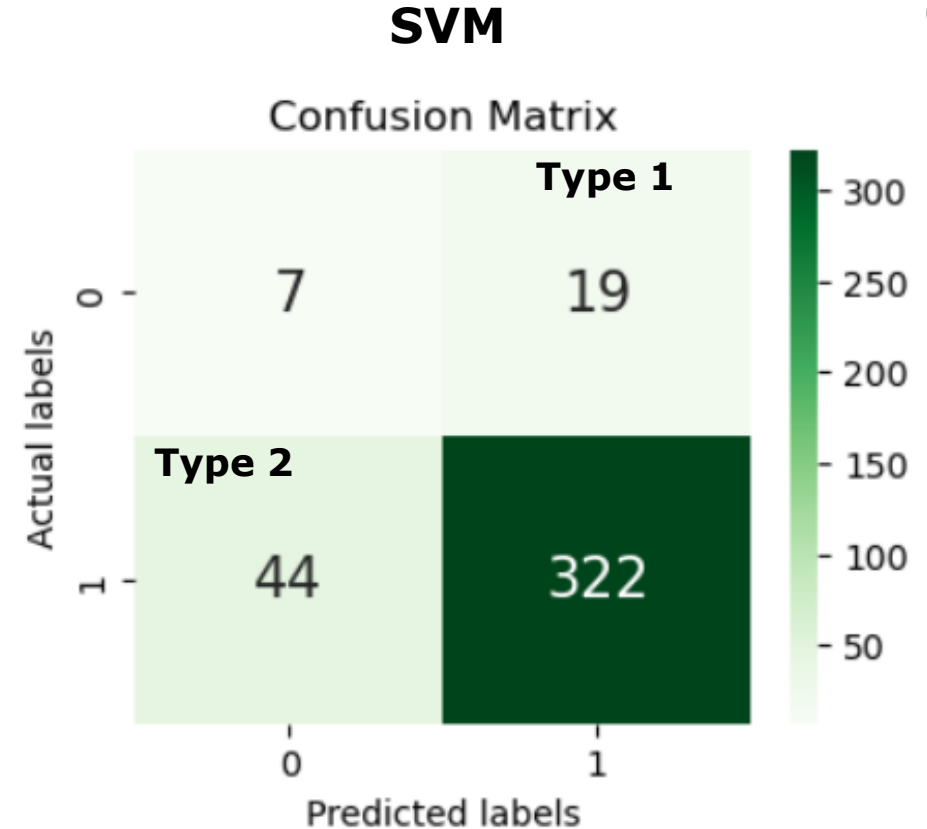
Confusion Matrix

- **Type Error 1 (FP)** Selling faulty Semiconductors-> is more costly = €1,000
- **Type Error 2 (FN)** Discarding functioning wafers = €250

1=Pass
0=Fail



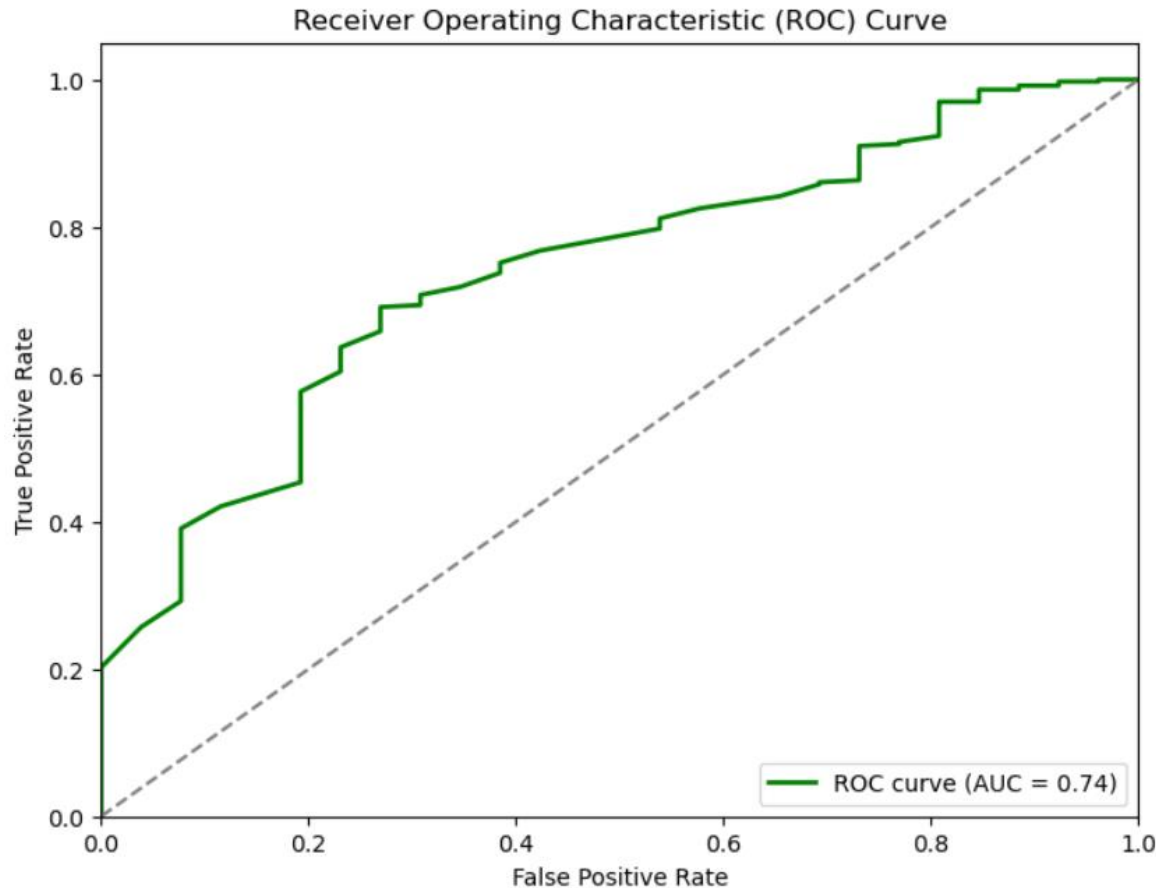
Total Cost= €26,000



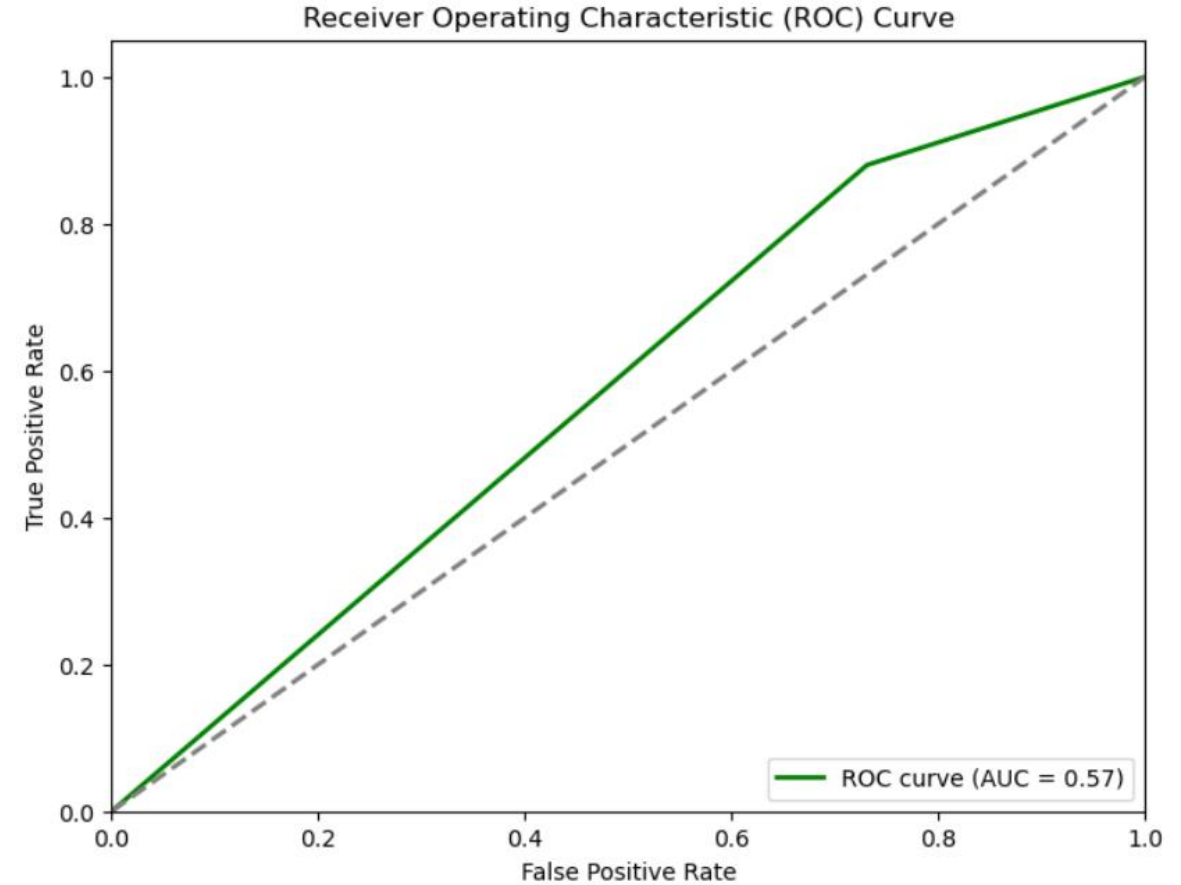
Total Cost= € 30,000

ROC Curve

Random Forest



SVM



The model has decent discriminatory power with an AUC of 0.74, indicating it performs better than a random classifier, but as seen in the Confusion Matrix the model needs to be reviewed.

Classification Report

Random Forest

	precision	recall	f1-score	support
0	0.20	0.19	0.20	26
1	0.94	0.95	0.94	366
accuracy			0.90	392
macro avg	0.57	0.57	0.57	392
weighted avg	0.89	0.90	0.89	392

SVM

	precision	recall	f1-score	support
0	0.14	0.27	0.18	26
1	0.94	0.88	0.91	366
accuracy			0.84	392
macro avg	0.54	0.57	0.55	392
weighted avg	0.89	0.84	0.86	392

Grid Search - Hyperparameter Tuning

Random Forest

```
# Set up the parameter grid to search
param_grid = {
    'n_estimators': [50, 100, 200],
    'max_depth': [None, 10, 20, 30],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 2, 4],
    'max_features': ['auto', 'sqrt', 'log2'],
    'bootstrap': [True, False]
}
```

Best Parameters: {'bootstrap': False, 'max_depth': 20, 'max_features': 'auto', 'min_samples_leaf': 1, 'min_samples_split': 5, 'n_estimators': 200}

Best parameters found: {'C': 100, 'gamma': 1, 'kernel': 'rbf'}

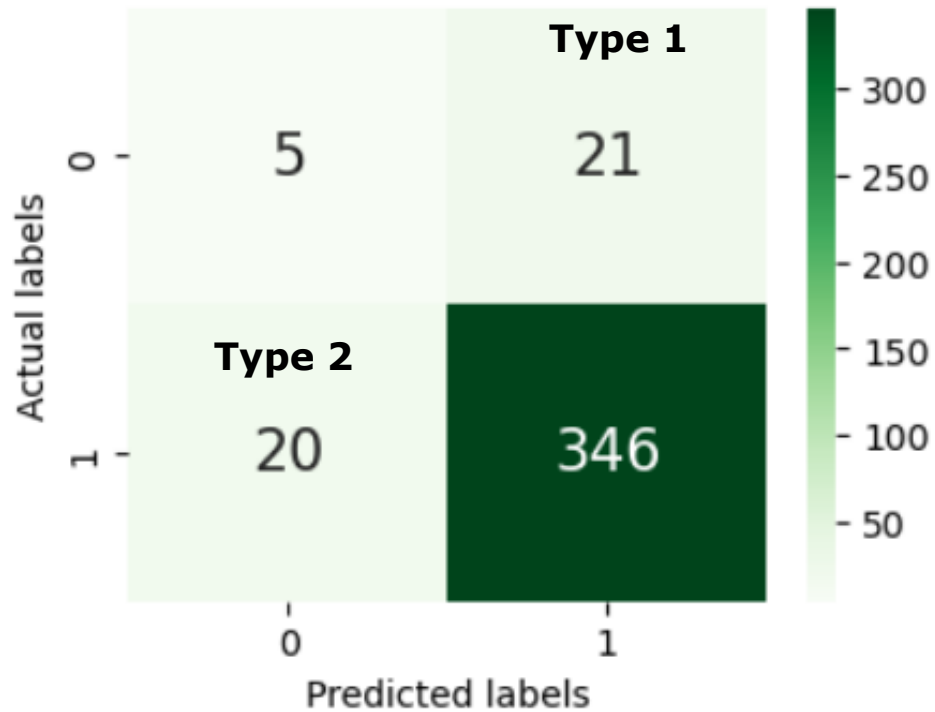
SVM

```
# Define the parameter grid
param_grid = {
    'C': [0.1, 1, 10, 100],
    'gamma': [1, 0.1, 0.01, 0.001],
    'kernel': ['linear', 'poly', 'rbf', 'sigmoid']
}
```

Grid Search - Confusion Matrix

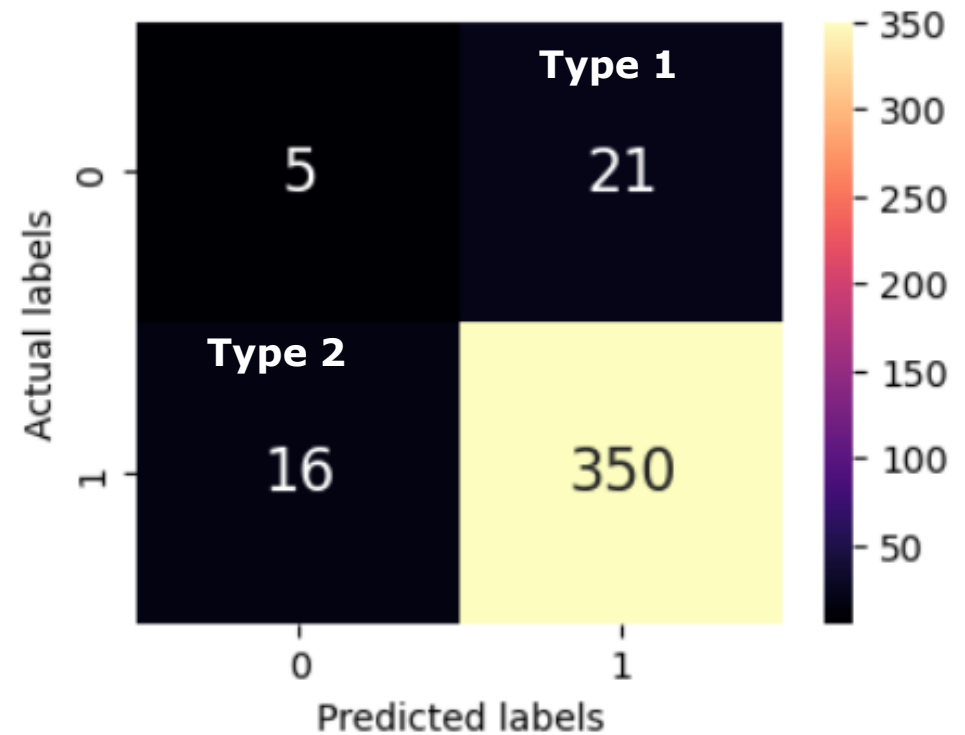
It did not improve a lot

Random Forest Before



Total Cost= €26,000

Random Forest After

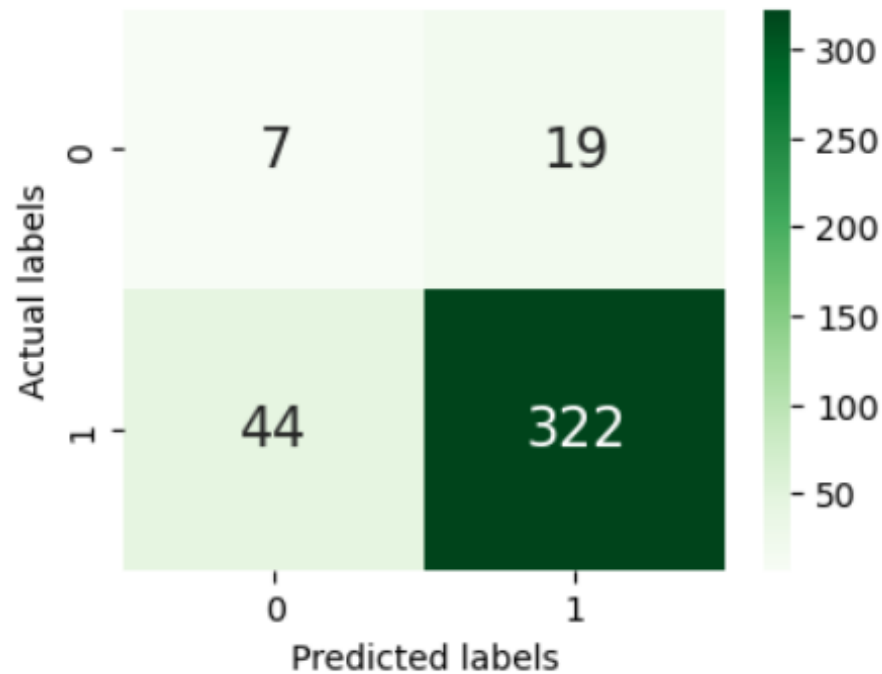


Total Cost= €25,000

Grid Search - Confusion Matrix

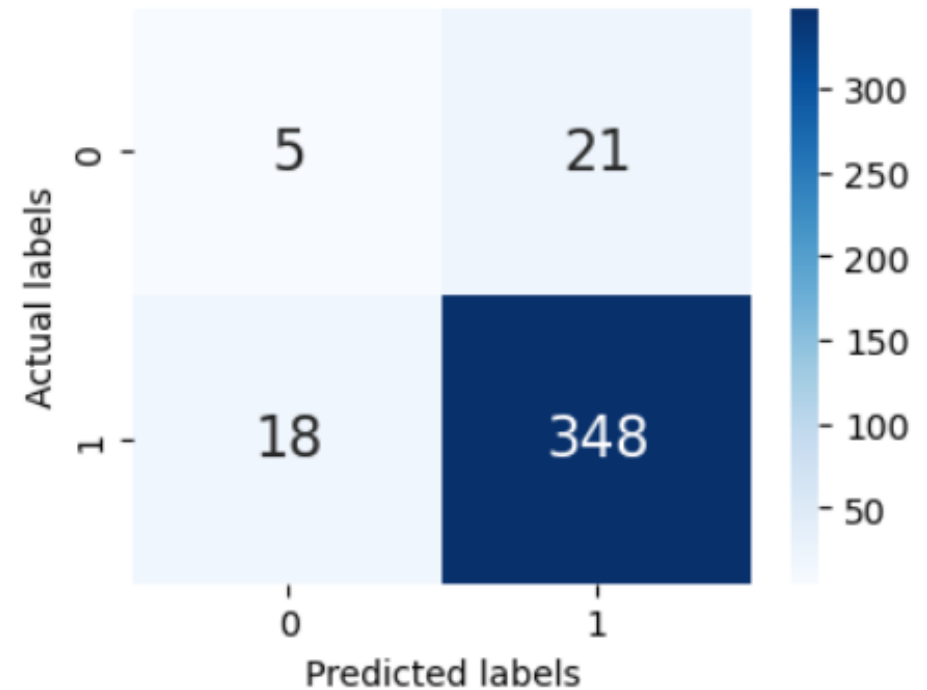
It did not improve a lot

SVM Before



Total Cost= €30,000

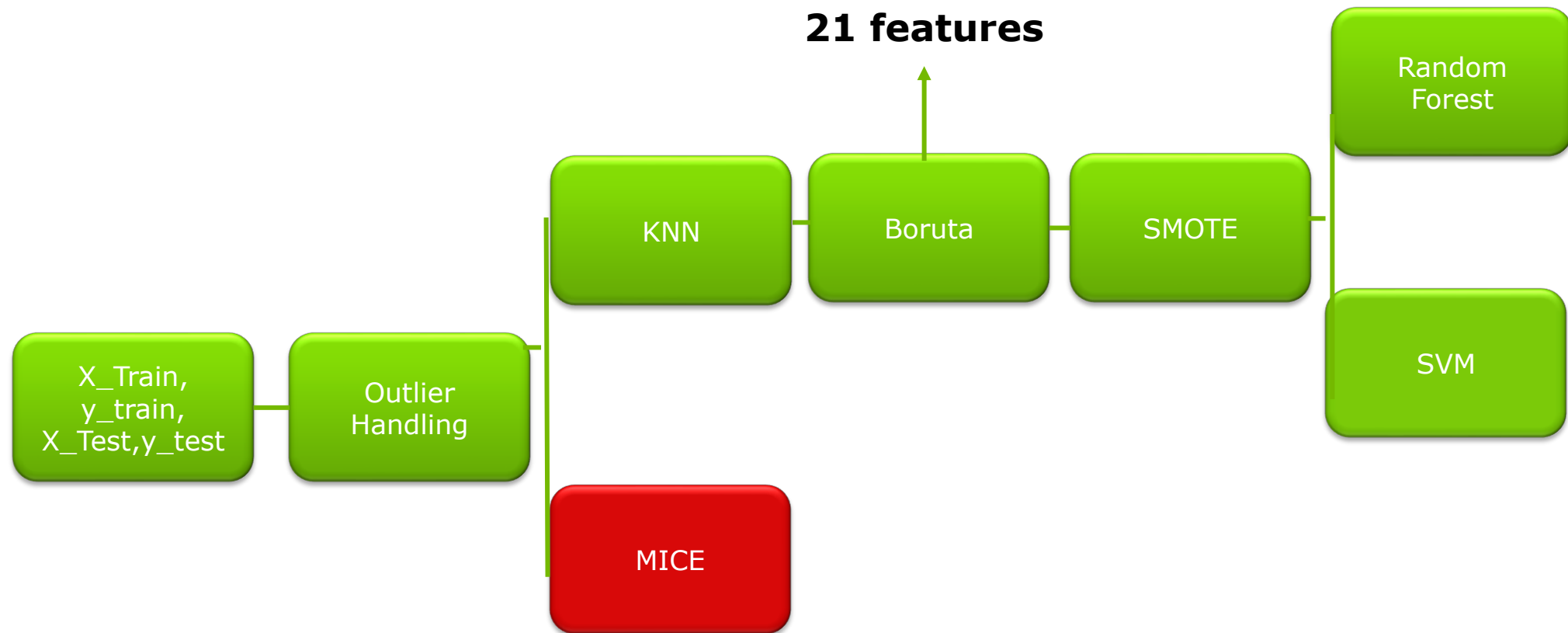
SVM After



Total Cost= €25,500

Model Tuning

Flow Chart Attempt 2

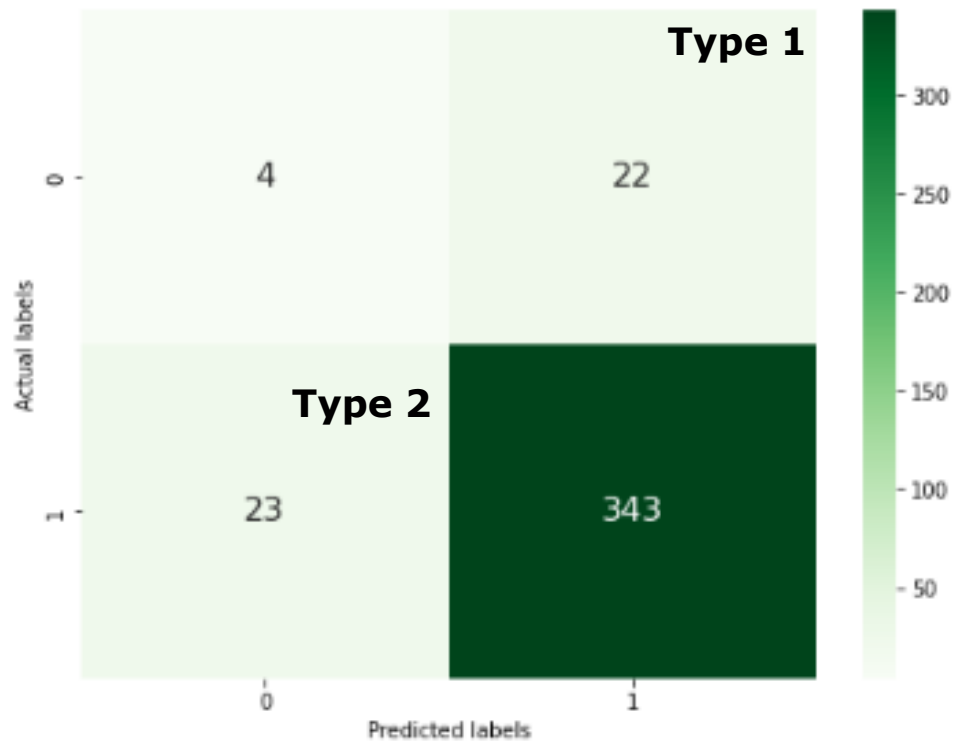


Confusion Matrix

- **Type Error 1** Selling faulty Semiconductors-> is more costly = €1,000
- **Type Error 2** Discarding functioning wafers = €250

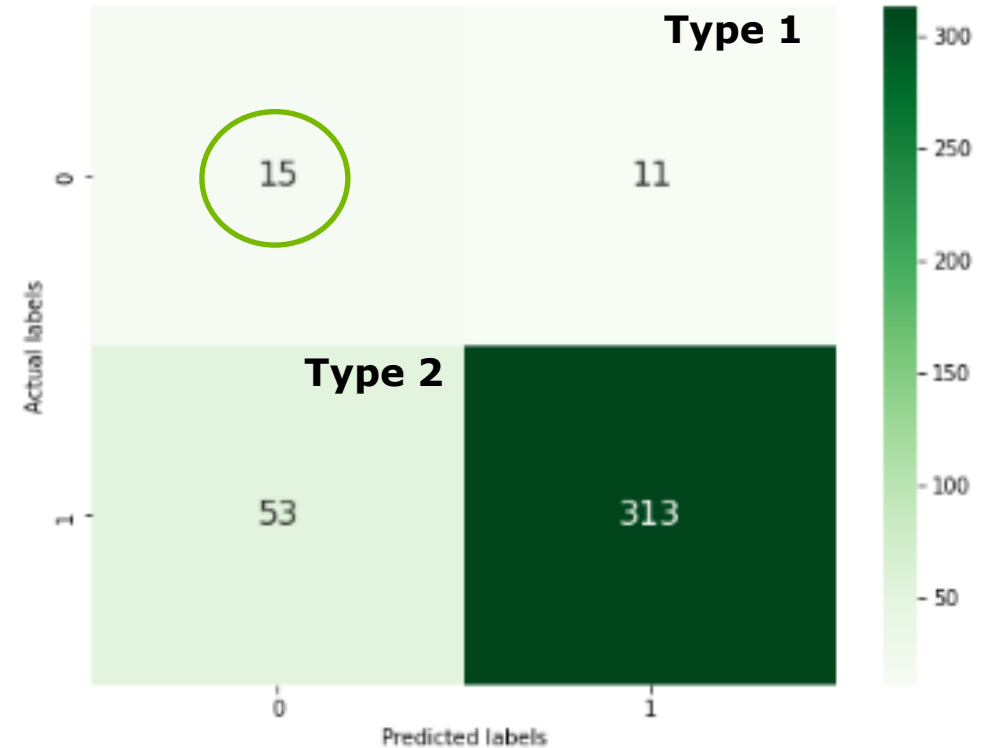
1=Pass
0=Fail

Random Forest



Total Cost= €27,750

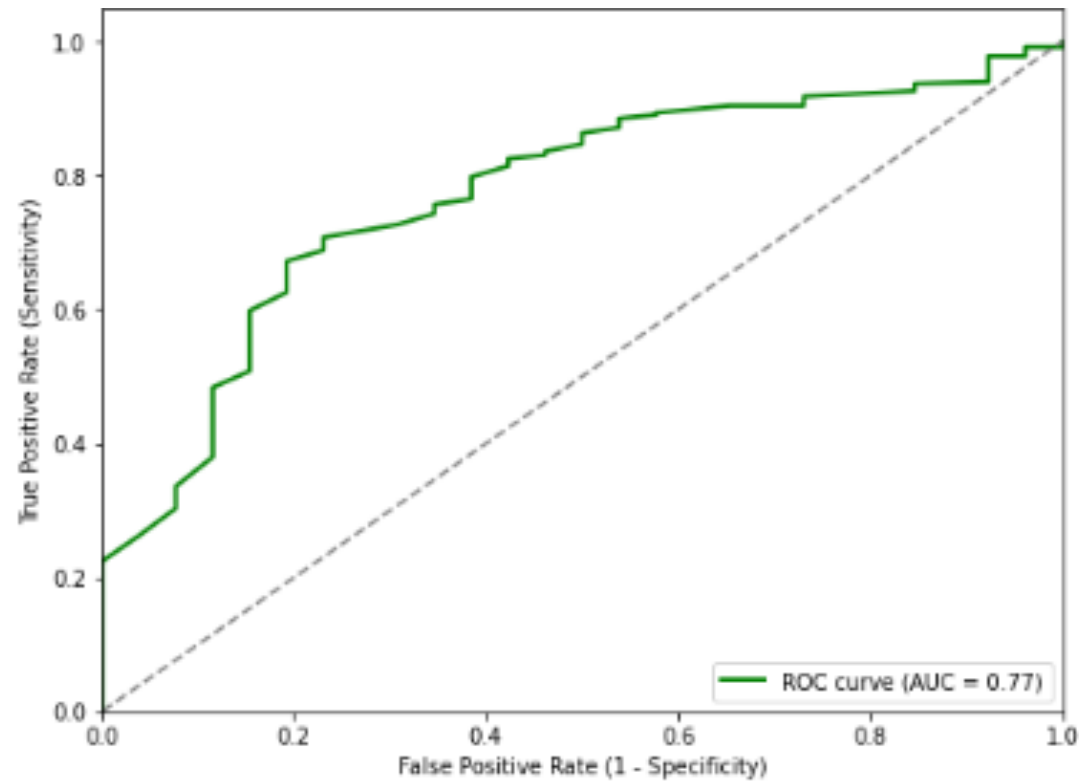
SVM



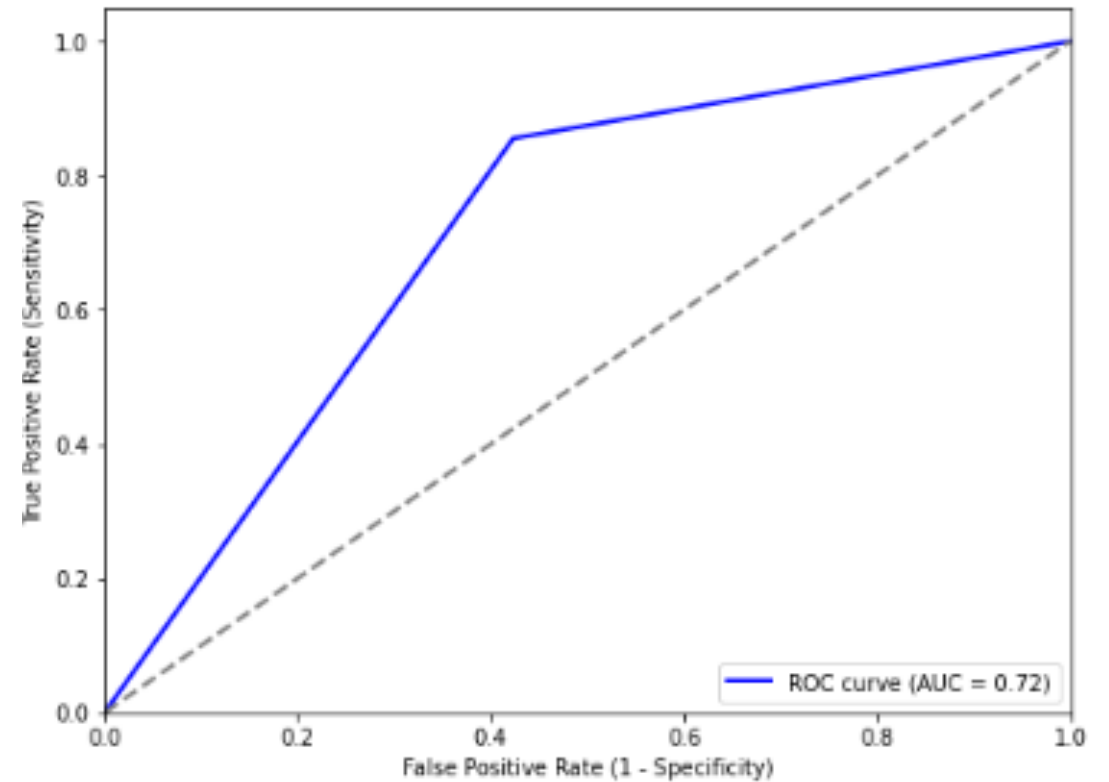
Total Cost= €24,250

ROC Curve

Random Forest



SVM



Classification Report

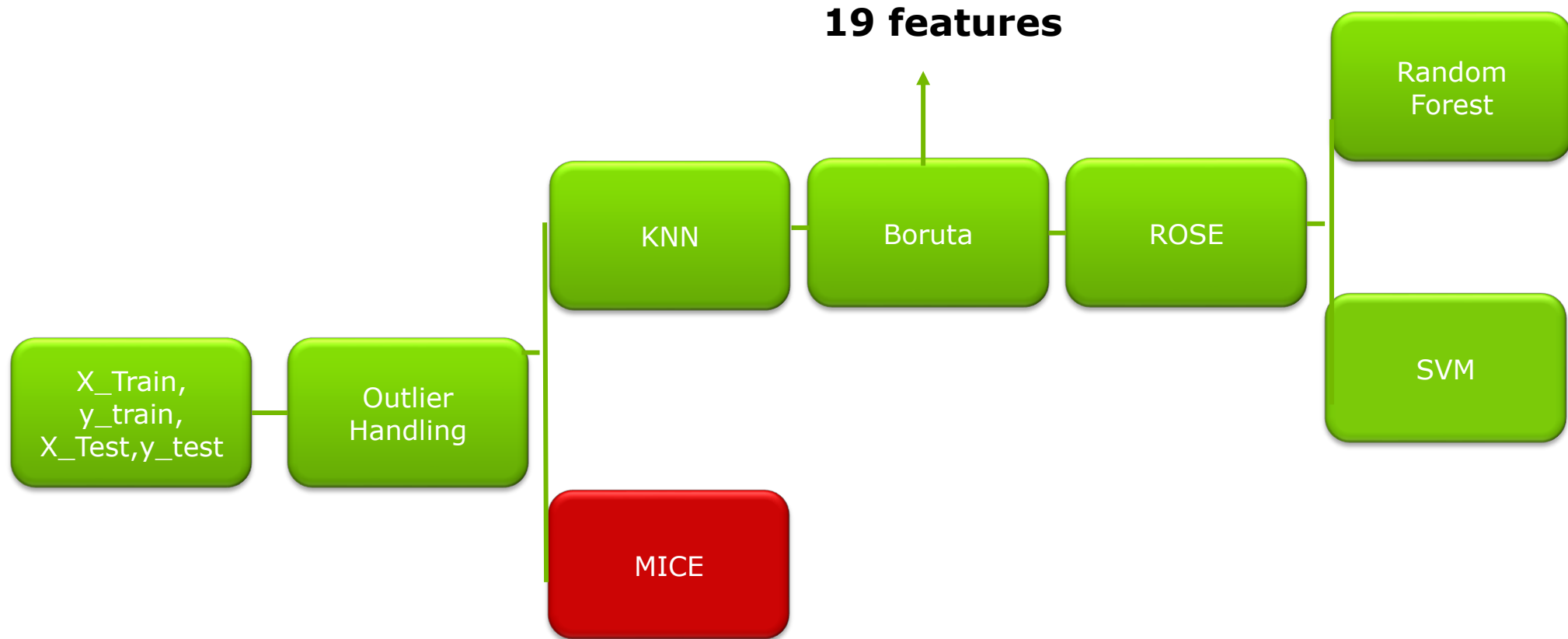
Random Forest

	precision	recall	f1-score	support
0	0.15	0.15	0.15	26
1	0.94	0.94	0.94	366
accuracy			0.89	392
macro avg	0.54	0.55	0.54	392
weighted avg	0.89	0.89	0.89	392

SVM

	precision	recall	f1-score	support
0	0.22	0.58	0.32	26
1	0.97	0.86	0.91	366
accuracy			0.84	392
macro avg	0.59	0.72	0.61	392
weighted avg	0.92	0.84	0.87	392

Flow Chart Attempt 3

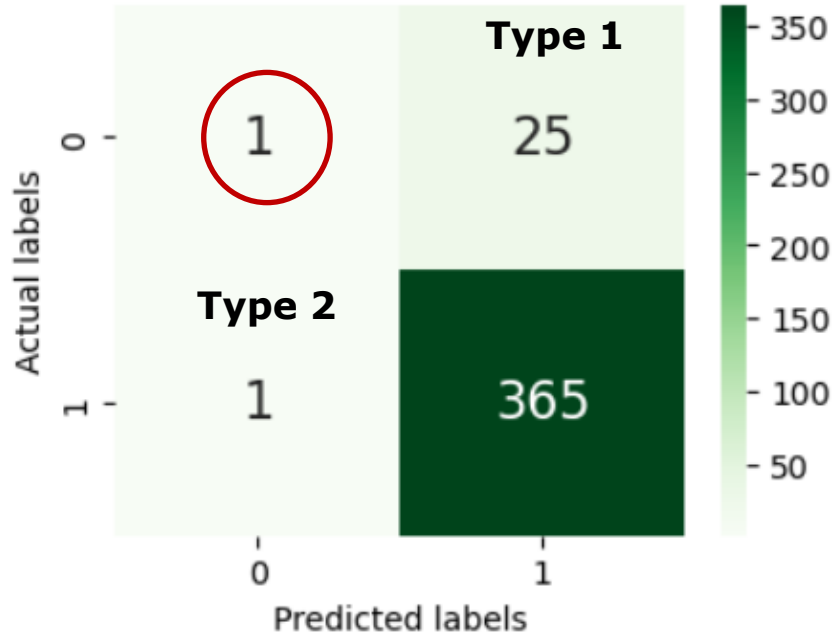


Confusion Matrix

- **Type Error 1** Selling faulty Semiconductors-> is more costly = €1,000
- **Type Error 2** Discarding functioning wafers = €250

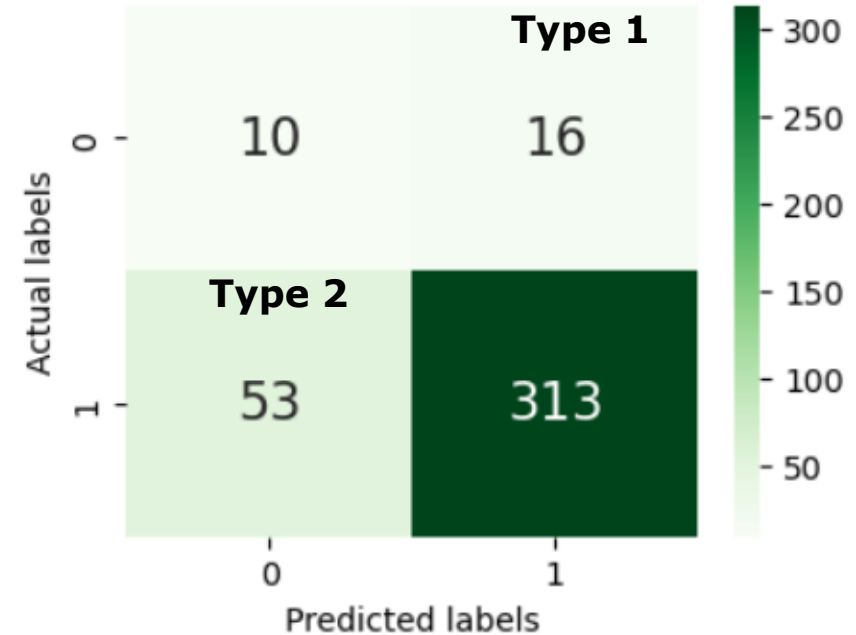
1=Pass
0=Fail

Random Forest



Total Cost= € 25,250

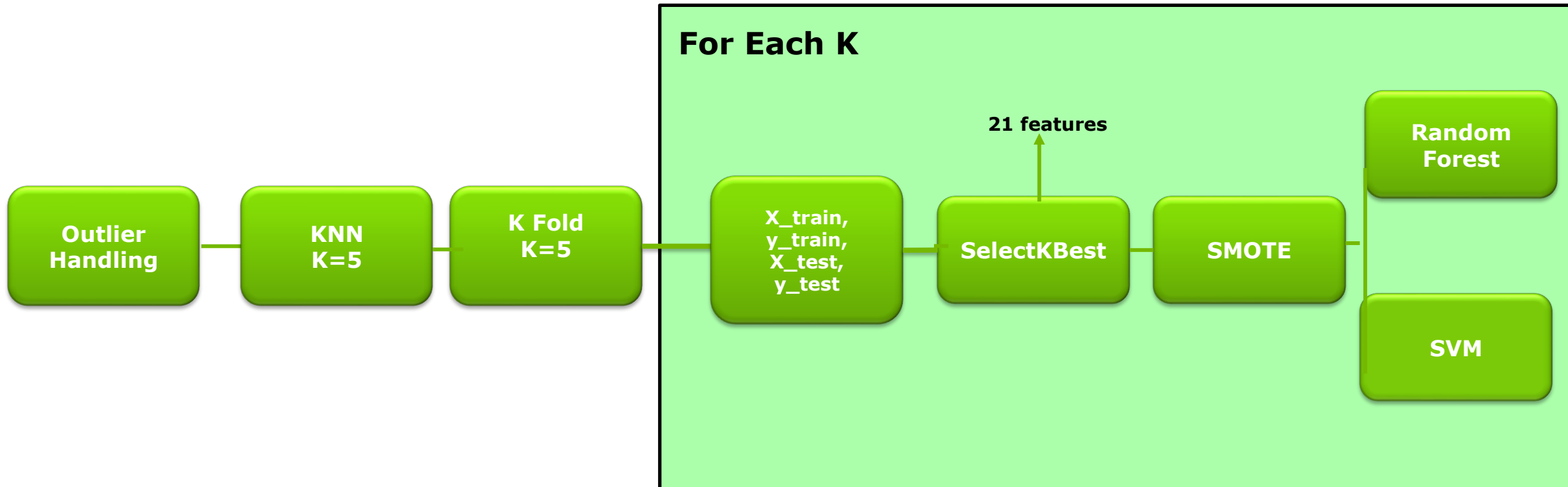
SVM



Total Cost= € 29,250

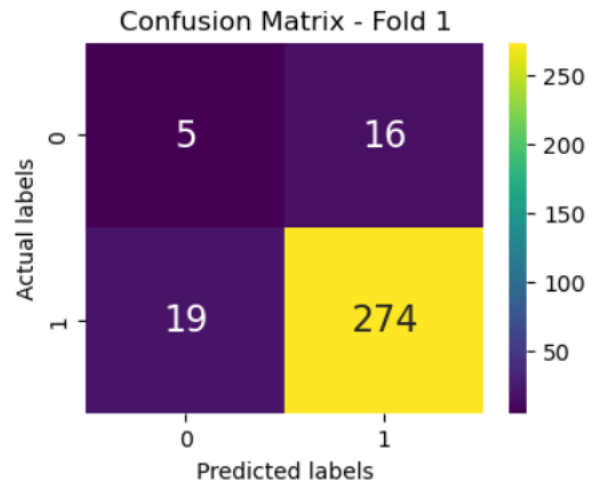
K Fold Approach

Flow Chart Stratified K-Fold

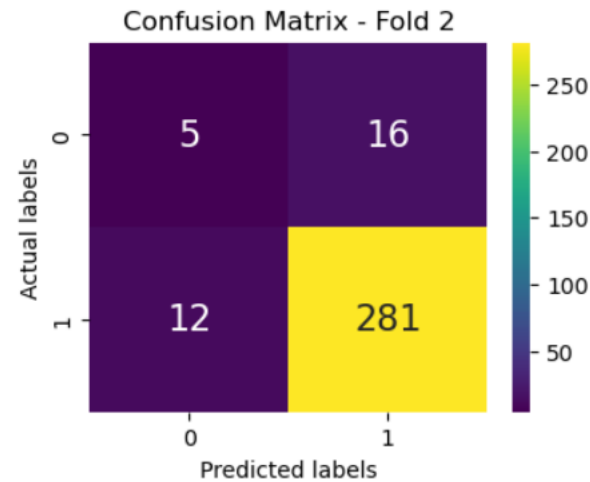


K-Fold Approach Random Forest

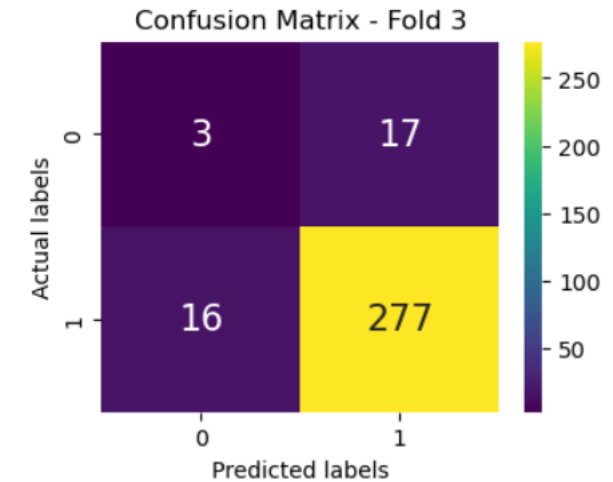
Fold 1 - Accuracy: 0.8885



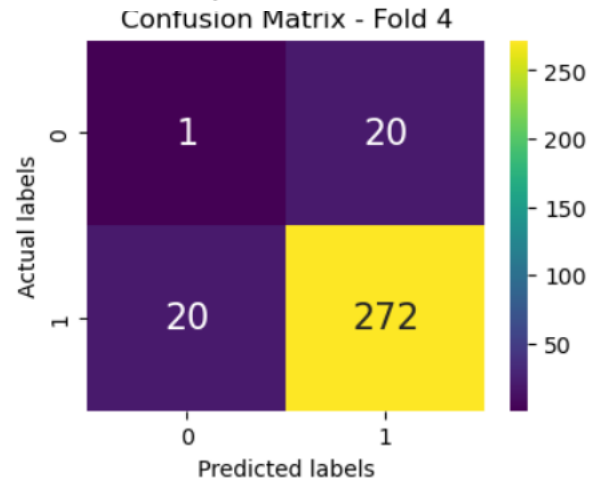
Fold 2 - Accuracy: 0.9108



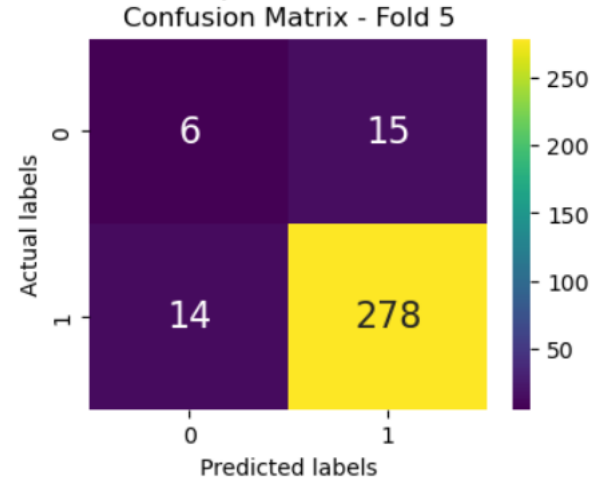
Fold 3 - Accuracy: 0.8946



Fold 4 - Accuracy: 0.8722



Fold 5 - Accuracy: 0.9073

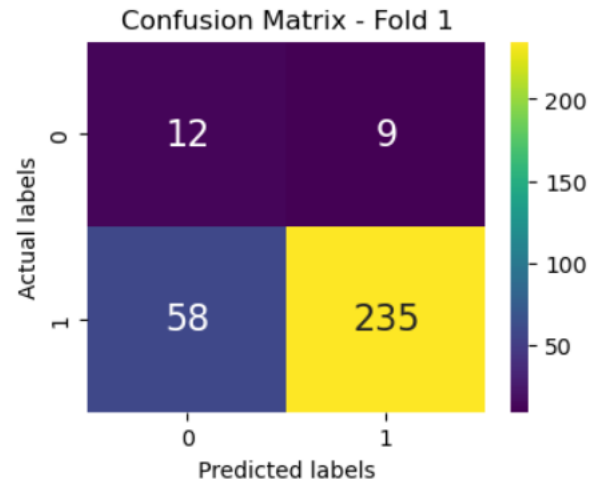


Average Accuracy: 0.8934

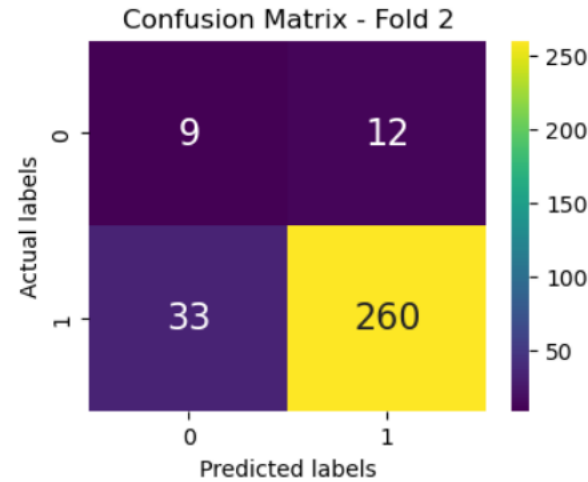
Volatility of
True
Negatives: 2

K-Fold Approach SVM

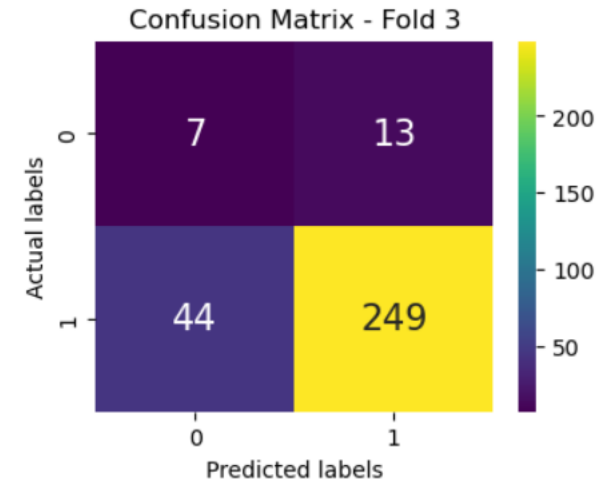
Fold 1 - Accuracy: 0.7866



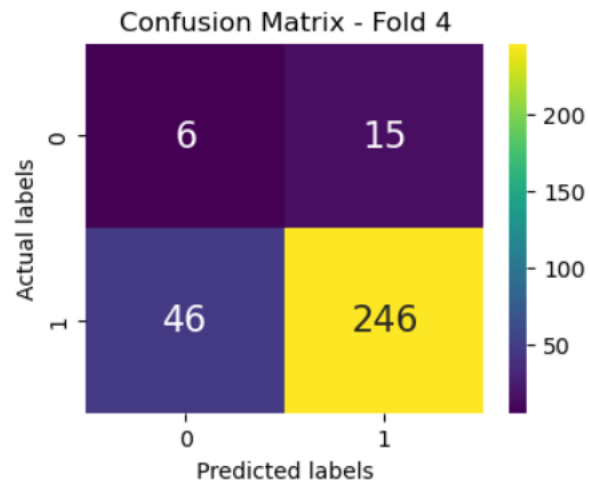
Fold 2 - Accuracy: 0.8567



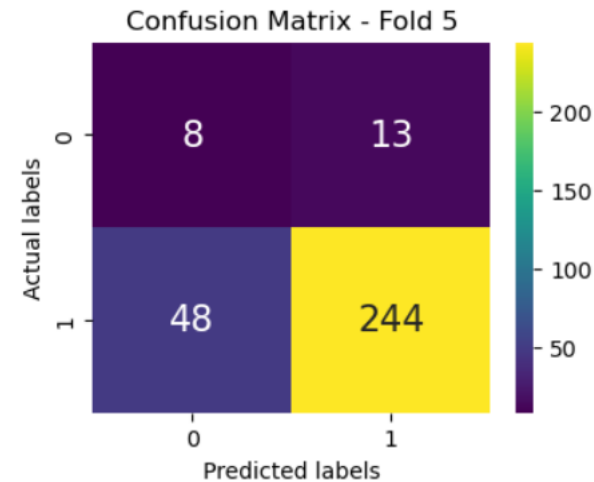
Fold 3 - Accuracy: 0.8179



Fold 4 - Accuracy: 0.8051



Fold 5 - Accuracy: 0.8051



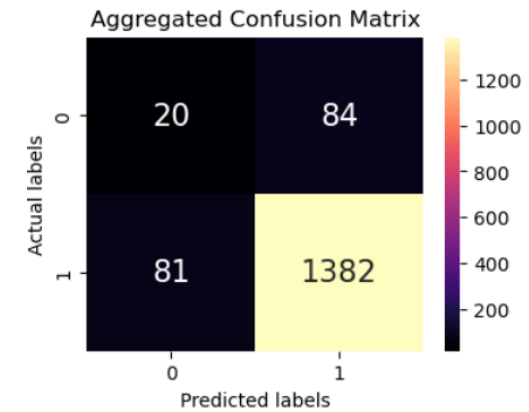
Average Accuracy: 0.8143

Volatility of
True
Negatives:
2.3

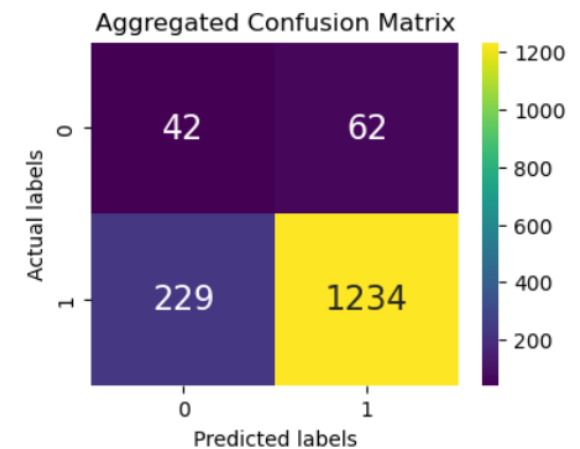
Comparison

	Random Forest	SVM
Average Precision	0.57	0.55
Average Recall	0.56	0.62
Average F1 score	0.56	0.56
Class 0 - Average Precision	0.20	0.16
Class 0 - Average Recall	0.19	0.40
Class 0 - Average F1 score	0.20	0.22
Class 1 - Average Precision	0.94	0.95
Class 1 - Average Recall	0.94	0.84
Class 1- Average F1 Score	0.94	0.89
Average Accuracy	0.89	0.81

Random Forest

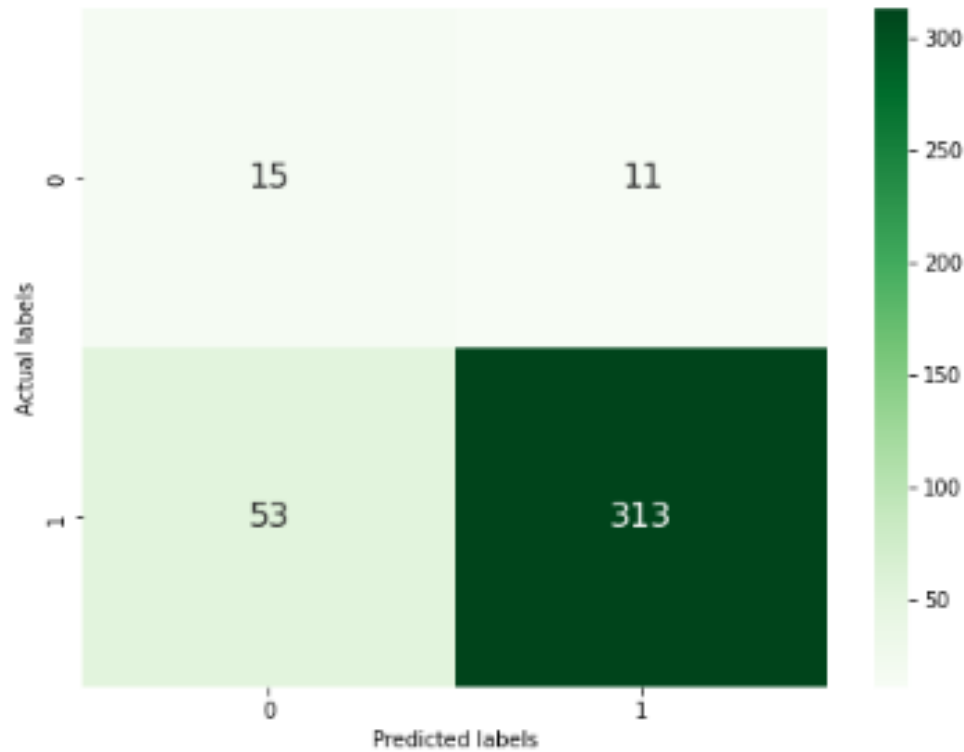


SVM

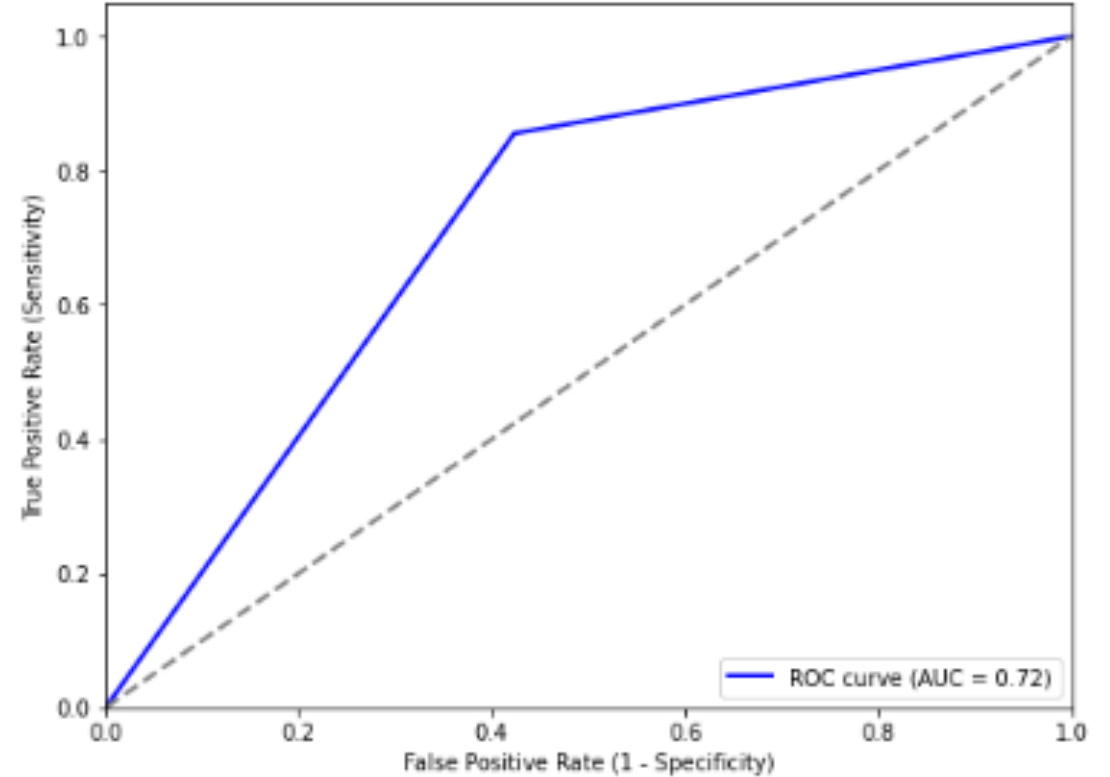


Model Selection

SVM



Total Cost= €24,250



	precision	recall	f1-score	support
0	0.22	0.58	0.32	26
1	0.97	0.86	0.91	366
accuracy			0.84	392
macro avg	0.59	0.72	0.61	392
weighted avg	0.92	0.84	0.87	392

Summary

1. We tried several approaches in the SECOM database.
 1. For imputation methods KNN and MICE.
 2. For feature selection BORUTA and SelectKBest
 3. For Balancing ROSE & SMOTE.
2. We created six models from features from BORUTA to try and get a Model which is more accurate than others.
3. We shortlisted Random Forest and SVM Model to go further with due to its goodness of fit measures.
4. To improve on these two model, We performed grid search on both.
5. We performed model tuning and different combinations to our two selected models to compare and reason to make it more accurate and cost effective.
6. We did Stratified K fold=5 to validate the volatility and performance metrics of our 2 best models.
7. We selected the best Model based on cost effectiveness.

Best Practices

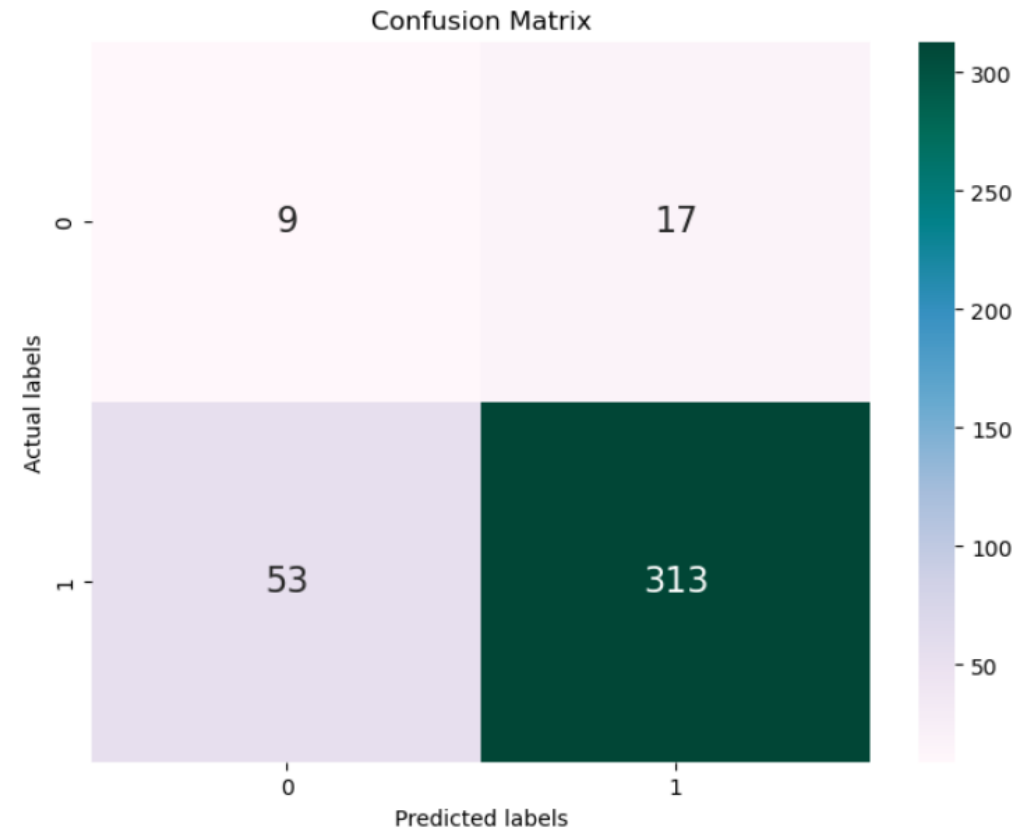
1. Understand your data thoroughly.
2. Determine correctly the Type 1 and Type 2 errors
3. Try different imputation method techniques depending on your model.
4. Have clear vision of when scaling is needed.
5. Visualize the data when you perform any substantial changes.
6. Understand what K fold does and when to use it.
7. ROC Curve+ Classification Report (F1 score).
8. Grid Search as an improvement tool.
9. Set relevant performance metrics.

Appendix

Modelling

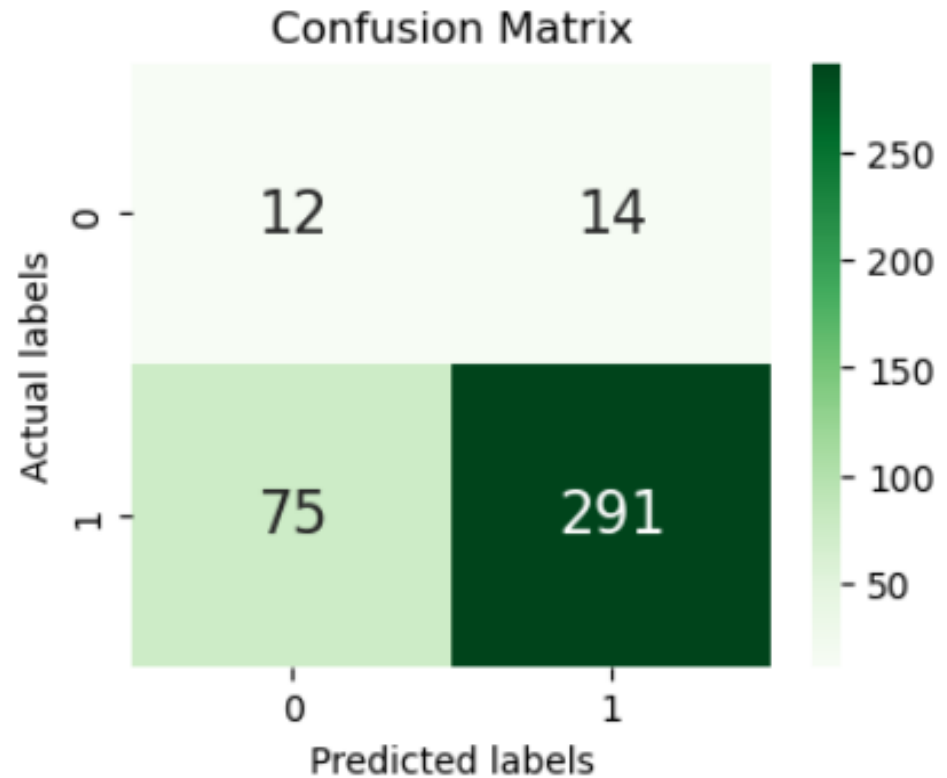
Decision Tree

Total Cost	\$30,250
F1 Score - 0	0.20
F1 Score - 1	0.90



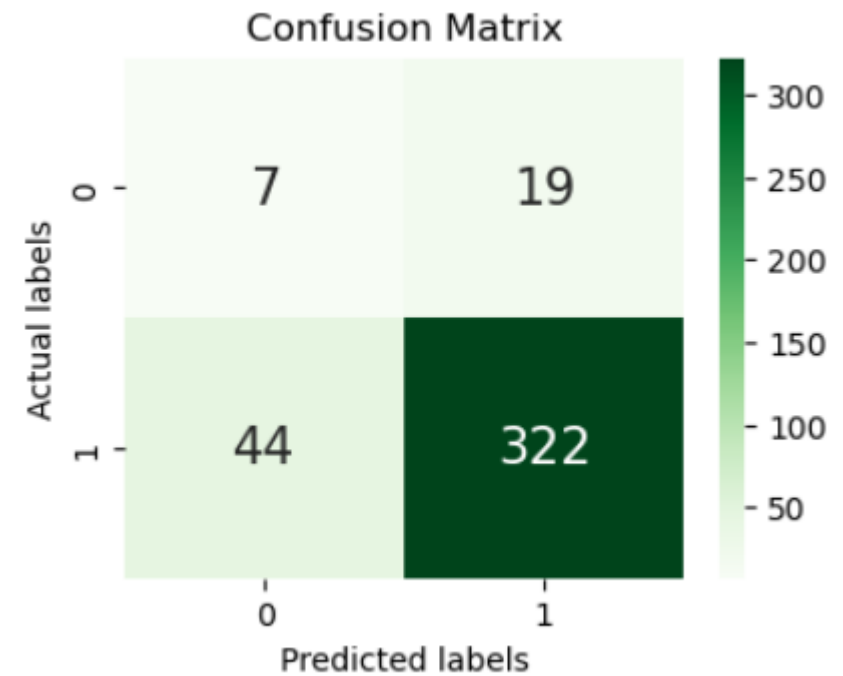
KNN Classifier – Neighbors 5

Total Cost	\$32,750
F1 Score - 0	0.17
F1 Score - 1	0.88



SVM

Total Cost	\$30,000
F1 Score - 0	0.18
F1 Score - 1	0.91



Gradient Boosting Classifier

Total Cost	\$30,500
F1 Score - 0	0.16
F1 Score - 1	0.91

